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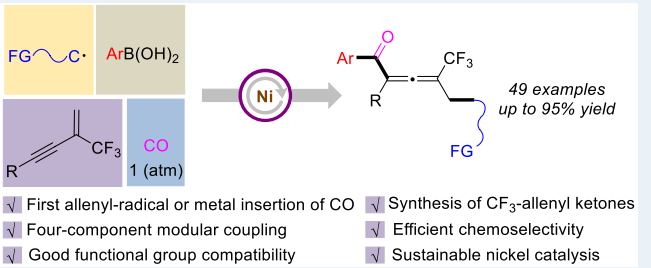
Nickel-Catalyzed Modular Four-Component 1,4-Alkylcarbonylation of 1,3-Enynes to Tetra-Substituted CF_3−Allenyl Ketones

Qi-Chao Shan, Yan Zhao, Shu-Tao Wang, Hong-Fei Liu, Xin-Hua Duan, and Li-Na Guo*

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ABSTRACT: The modular four-component carbonylation of unsaturated hydrocarbons represents an elegant strategy for the one-pot synthesis of complex carbonyl compounds. However, this strategy is currently focused on the 1,2-difunctionalization of olefins or alkynes, and the 1,4-alkylcarbonylation of 1,3-enynes remains unexplored. In this study, we report a Ni-catalyzed 1,4-alkylcarbonylation of CF_3-containing 1,3-enynes under 1 atm of CO, which affords the tetra-substituted CF_3-allenyl ketones with good yields and selectivity. This protocol features mild conditions, broad substrate scope, and acceptable functional group compatibility. Control experiments revealed that the reactivity of oxime esters (regulated by leaving groups) and polarity matching are crucial for the success of this four-component cascade. Preliminary mechanistic studies suggest that the acyl nickel complex is a key intermediate in this transformation.



KEYWORDS: *1,4-alkylcarbonylation, nickel catalysis, modular synthesis, alkyl radical, four-component coupling*

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In modern organic synthesis, multicomponent modular synthesis represents the simplest and most elegant strategy for the one-step construction of high-value fine chemicals from cheap bulk chemicals.[1] The widespread occurrence of carbonyl skeletons in natural products, pharmaceuticals, and polymers stimulates chemists to develop carbonylative multicomponent reactions (CMCRs) to access diverse and complex carbonyl compounds from simple and available starting materials.[2] In recent years, Wu, Ryu, Beller, and others have reported several elegant transition-metal (Pd, Co, and Cu) catalyzed, radical-mediated four-component 1,2-dicarbofunctionalizations of olefins for the synthesis of ketones, esters, and amides (Scheme 1a).[3] In contrast, nickel-catalyzed carbonylative multicomponent reactions (CMCRs) are much less developed, although Ni catalysis exhibits excellent catalytic properties in various cross-coupling reactions. This is probably due to the formation of highly toxic $Ni(CO)_4$ and $Ni(CO)_3L$.[4] On the other hand, the strong binding affinity of CO to nickel blocks the interaction between nickel and substrates. In 2019, Zhang's group achieved the first Ni-catalyzed carbonylation couplings of difluoroalkyl bromides with arylboronic acids, which greatly enriched the radical-mediated carbonylation.[4b] After that, Zhang's group successfully extended the Ni catalysis to the four-component carbonylation coupling of alkenes.[5] In addition, Wu's group[6] also developed the nickel-catalyzed four-component 1,2-carbocarbonylation of olefins with amines or alcohols. The atmospheric pressure of CO and the broad nucleophile scope merits made Ni catalysis more attractive than other metal catalysts. Overall, the four-component carbonylation reactions have focused on the 1,2-difunctionalization of olefins involving fluorinated alkyl radicals (•CF_3 or •CF_2COR) and alkoxyalkyl radicals, producing the monocarbonyl compounds. In this regard, four-component carbonylation reactions involving the alkyl radicals are less reported, which may be due to the undesired β-hydrogen elimination, radical polarity mismatch, and reaction rate incompatibility. Therefore, it is still highly desirable to develop new modular four-component carbonylative reactions with inexpensive transition-metal catalysis and various unsaturated systems as well as radical precursors under atmospheric pressure of CO.

1,3-Enynes are important building blocks in organic synthesis, which can serve as the efficient radical acceptors to undergo 1,2- and 1,4-difunctionalizations.[7] Among them, Wang's group[7d] reported a nickel-catalyzed radical 1,4-dicarbofunctionalization of 1,3-enynes to the fluoroalkylated allenes, in which the key allenyl-Ni^{III} intermediate was formed by the interaction of Ni^{II} species with the allenyl radical (generated through the tautomerization of the propargyl radical). Later on, Kong[7h] reported another 1,4-dicarbofunctionalization of 1,3-enynes through the combination of decatungstate photo-HAT and nickel catalysis. There, a 1,3-Ni shift pathway was proposed. We envision whether the 1,4-

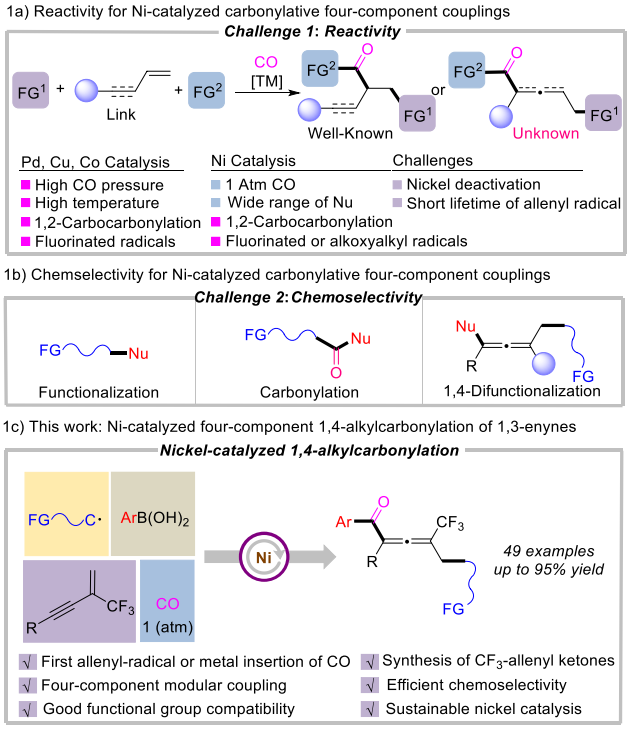
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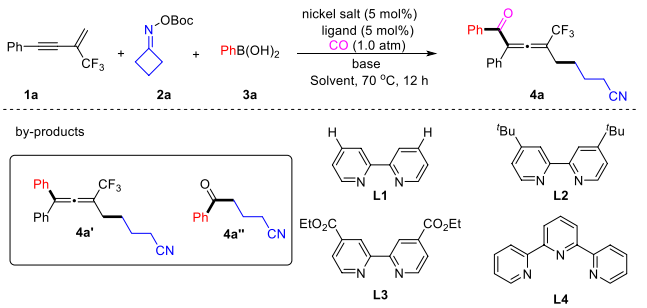
Scheme 1. Challenges of the Ni-Catalyzed Four-Component 1,4-Alkylcarbonylation of 1,3-Enynes



alkylcarbonylation of 1,3-enynes can be achieved with Ni catalysis under a CO atmosphere (Scheme 1c). One challenge of this carbonylation reaction is the reactivity problem, including the possible deactivation of Ni species in the presence of CO and the short lifetime of allenyl radicals.[8] Another major challenge of this four-component reaction is the selectivity problem, because there are several competing side reactions involved, including the direct coupling of alkyl radicals with nucleophiles,[9] three-component carbonylative coupling,[10] and three-component coupling without carbonylation[11] (Scheme 1b). Therefore, it is very critical to regulate the matching performance of the reactivity and selectivity to achieve this multicomponent carbonylation reaction.

Considering the polarity matching, we elected the CF_3-containing 1,3-enyne **1a** as the radical acceptor, cyclobutanone oxime ester[12] **2a** as the alkyl radical precursor, and phenylboronic acid **3a** as the nucleophile to optimize the reaction conditions with Ni catalysis (Table 1). After many trials, it was found that the four-component 1,4-alkylcarbonylation of 1,3-enyne proceeded efficiently to give the desired tetrasubstituted CF_3-allenyl ketone **4a** under Ni catalysis in the presence of 1 atm of CO pressure. Screening of different nickel salts showed that $Ni(PPh_3)_2Br_2$ had the best catalytic efficiency, giving expected allenyl ketone **4a** in 83% yield (entries 1−4). A number of solvents were then tested, and 1,4-dioxane was found to be the best one (entries 5−7). It is worth mentioning that using MTBA as a solvent, the three-component product **4a′** without CO insertion was obtained as a single product (entry 5). Various organic and inorganic bases such as DIPEA and K_2CO_3 were investigated, and both gave lower yields than Et_3N (entries 8 and 9). To our delight, the yield of **4a** was improved to 91% when 5.0 equiv of H_2O was added, probably due to the promotion of transmetalation (entry 10).[5a,13] The survey of ligands indicated that the

Table 1. Optimization of Reaction Conditions[a]



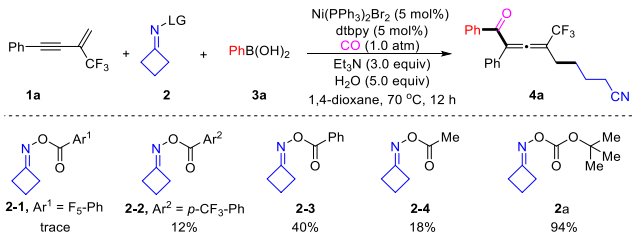
entry	nickel salt	solvent	base	L	4a/4a′/4a′′ yield (%)[b]
1	$Ni(cod)_2$	1,4-dioxane	Et_3N	L2	62/n.d./trace
2	$Ni(PPh_3)_2Cl_2$	1,4-dioxane	Et_3N	L2	75/n.d./8
3	$NiBr_2$·Glyme	1,4-dioxane	Et_3N	L2	63/n.d./trace
4	$Ni(PPh_3)_2Br_2$	1,4-dioxane	Et_3N	L2	83/n.d./trace
5	$Ni(PPh_3)_2Br_2$	MTBA	Et_3N	L2	n.d./54/n.d.
6	$Ni(PPh_3)_2B_2$	THF	Et_3N	L2	61/n.d./trace
7	$Ni(PPh_3)_2Br_2$	EtOAc	Et_3N	L2	25/n.d./trace
8	$Ni(PPh_3)_2Br_2$	1,4-dioxane	DIPEA	L2	38/n.d./trace
9	$Ni(PPh_3)_2Br_2$	1,4-dioxane	K_2CO_3	L2	73/n.d./trace
10[c]	$Ni(PPh_3)_2Br_2$	1,4-dioxane	Et_3N	L2	91/n.d./trace
11[c]	$Ni(PPh_3)_2Br_2$	1,4-dioxane	Et_3N	L1	63/n.d./trace
12[c]	$Ni(PPh_3)_2Br_2$	1,4-dioxane	Et_3N	L3	50/n.d./trace
13[c]	$Ni(PPh_3)_2Br_2$	1,4-dioxane	Et_3N	L4	trace/n.d./n.d.
14[c,d]	$Ni(PPh_3)_2Br_2$	1,4-dioxane	Et_3N	L2	94/n.d./trace
15[e]	Without CO				n.d./45/n.d.
16	Without **1a**				n.d./n.d./66

[a]Reaction conditions: **1a** (0.1 mmol, 1.0 equiv), **2a** (0.2 mmol, 2.0 equiv), **3a** (0.2 mmol, 2.0 equiv), nickel salt (5 mol %), ligand (5 mol %), and base in a solvent (1.5 mL) under 1.0 atm of CO at 70 °C for 12 h. [b]Isolated yield. [c]5.0 equiv of H_2O was used. [d]3.0 equiv of Et_3N was used. [e]**1a** (0.1 mmol, 1.0 equiv), **2** (0.2 mmol, 2.0 equiv), **3a** (0.2 mmol, 2.0 equiv), $Ni(PPh_3)_2Br_2$ (5 mol %), dtbpy (5 mol %), Et_3N (3.0 equiv), and H_2O (5.0 equiv) in 1,4-dioxane (1.5 mL) under N_2 at 70 °C for 12 h. L = ligand, n.d. = not detected.

dipyridine-type ligands are uniquely effective (entries 11−13). Increasing the amount of Et_3N to 3.0 equiv resulted in a slightly higher yield of 94% of **4a** (entry 14). Finally, control experiments were performed. In the absence of CO gas, the three-component 1,4-alkylarylation of 1,3-enyne occurred to give allene **4a′** in 45% yield (entry 15). In the absence of 1,3-enyne **1a**, the arylcarbonylation of cyclobutanone oxime ester proceeded smoothly to give ketone **4′′** in 66% yield (entry 16). These results highlighted the excellent chemoselectivity of this nickel catalyzed multicomponent cascade. Further investigation revealed that the nickel catalyst, ligand, and base were all essential to the success of this four-component reaction (not shown). It is known that the leaving groups in oxime esters usually have significant influence on the reaction efficiency. Therefore, the cyclobutanone oxime esters with different leaving groups were evaluated (Scheme 2). Replacement of Boc with F_5−PhCO, *p*-CF_3−PhCO, PhCO, and MeCO resulted in poor yields of **4a** accompanied by the decomposition of most of the 1,3-enyne. We speculate that the rate of alkyl radical formation is likely to be different for cycloketone oxime esters with different leaving groups and that rate matching is essential for four-component carbonylation coupling.

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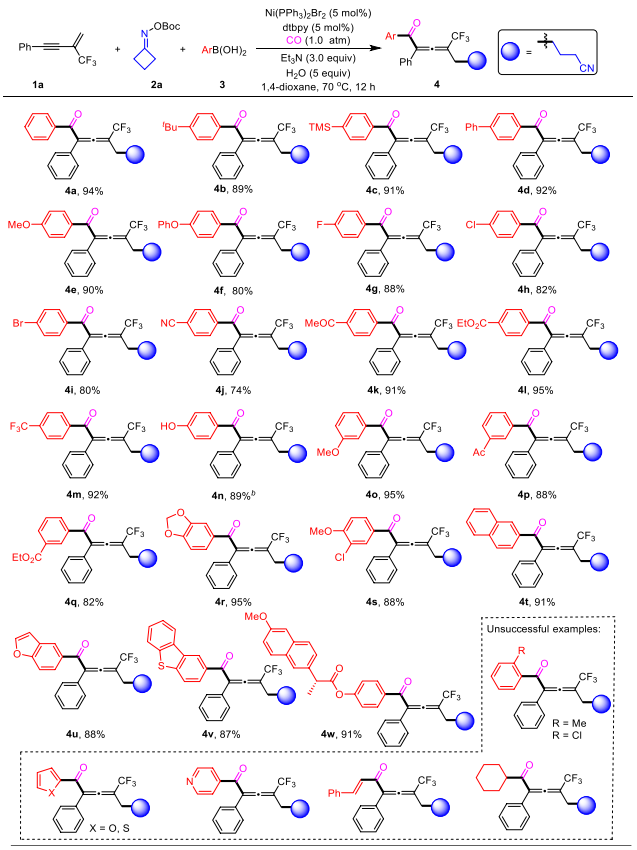
Scheme 2. Screening of Leaving Groups[a]



[a]Reaction conditions: **1a** (0.1 mmol, 1.0 equiv), **2** (0.2 mmol, 2.0 equiv), **3a** (0.2 mmol, 2.0 equiv), $Ni(PPh_3)_2Br_2$ (5 mol %), dtbpy (5 mol %), Et_3N (3.0 equiv), and H_2O (5.0 equiv) in 1,4-dioxane (1.5 mL) under 1.0 atm of CO at 70 °C for 12 h.

With the optimal conditions in hand, the generality and limitations of various arylboronic acids were first investigated using 1,3-enyne **1a** and oxime ester **2a** as the coupling partners (Scheme 3). Arylboronic acids with electron-withdrawing

Scheme 3. Scope of Arylboronic Acids[a]

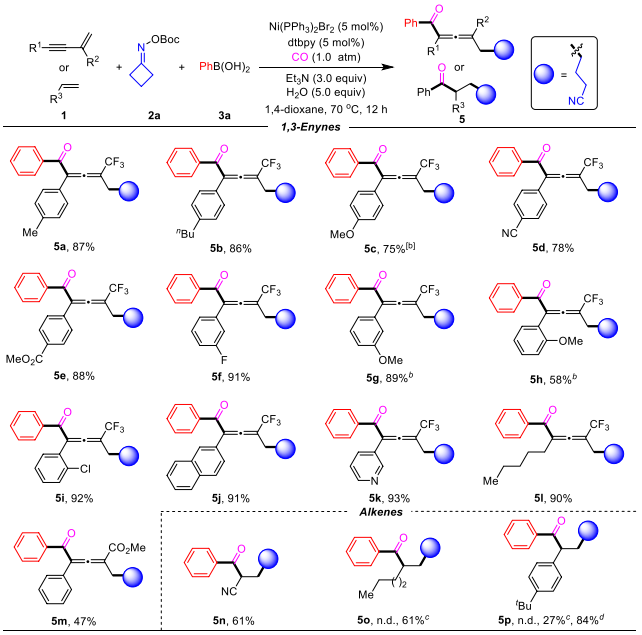


[a]Reaction conditions: **1a** (0.1 mmol, 1.0 equiv), **2a** (0.2 mmol, 2.0 equiv), **3** (0.2 mmol, 2.0 equiv), $Ni(PPh_3)_2Br_2$ (5 mol %), dtbpy (5 mol %), Et_3N (0.3 mmol, 3.0 equiv), and H_2O (0.5 mmol, 5.0 equiv) in 1,4-dioxane (1.5 mL) under 1.0 atm of CO at 70 °C for 12 h. [b]75 °C.

groups or electron-donating groups in the *para* position of the aromatic ring all reacted well, giving the desired four-component coupling products **4a**-**4n** in moderate to excellent yields. A variety of useful functional groups including trimethylsilyl (**4c**), bromine (**4i**), cyan (**4j**), acetyl (**4k**), ester (**4l**), and trifluoromethyl (**4m**) survived well under this nickel catalyst system with satisfactory yields ranging from 74% to 95%. Remarkably, 4-hydroxyphenylboronic acid was also amenable, furnishing desired allenyl ketone **4n** in 89% yield. The *meta*- and multisubstituted arylboronic acids were found to be the suitable substrates for this transformation (**4o**-**4s**). However, the *ortho*-substituted substrates failed to give any desired products, probably due to the steric hindrance effect. Fortunately, the heterarylboronic acids containing benzofuran and dibenzothiophene participated well in this reaction, giving the desired products **4u** and **4v** in 88% and 87% yields, respectively. The arylboronic acid masked with naproxen also gave desired product **4w** in 91% yield.

Next, the scope of 1,3-enynes for this nickel-catalyzed carbonylative multicomponent reaction was investigated. As shown in Scheme 4, a variety of CF_3-containing 1,3-enynes

Scheme 4. Scope of 1,3-Enynes and Alkenes[a]



[a]Reaction conditions: **1** (0.1 mmol, 1.0 equiv), **2a** (0.2 mmol, 2.0 equiv), **3a** (0.2 mmol, 2.0 equiv), $Ni(PPh_3)_2Br_2$ (5 mol %), dtbpy (5 mol %), Et_3N (0.3 mmol, 3.0 equiv), and H_2O (0.5 mmol, 5.0 equiv) in 1,4-dioxane (1.5 mL) under 1.0 atm of CO at 70 °C for 12 h. [b]65 °C. [c]Yield of **4a′′**. [d]Recovery of alkene.

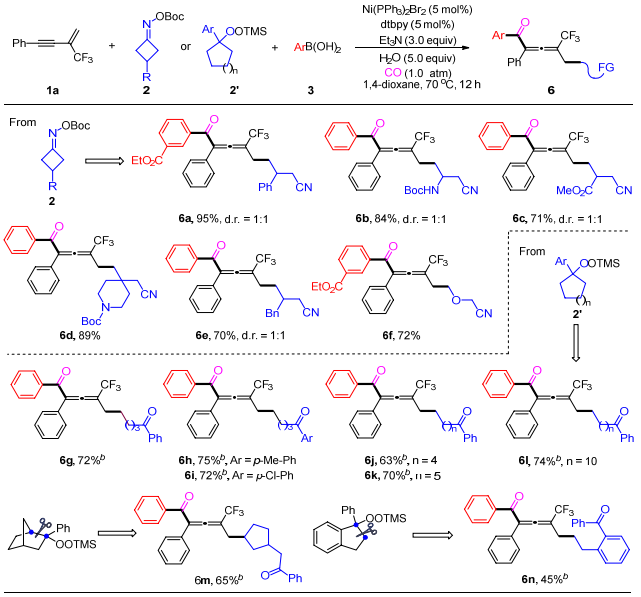
bearing a wide range of functional groups such as cyano (**5d**), ester (**5e**), and halogen (**5f** and **5i**) on the aromatic ring of the alkyne moiety delivered the corresponding CF_3-allenyl ketones in moderate to good yields. The electronic and steric effects of the alkyne moiety did not show any significant effect on the reaction efficiency. The 2-naphthyl (**5j**) and pyridyl (**5k**) 1,3-enynes also afforded the desired products in excellent yields. Notably, 1,3-enyne derived from an aliphatic alkyne also worked well to give the target product **5l** in 90% yield. In addition to the CF_3 group, ester-modified 1,3-enyne also provided 47% yield of the target product **5m**. To verify the compatibility of this nickel catalysis with other coupling partners, a range of alkenes was investigated to realize the 1,2-alkylcarbonylation. As we expected, an electron-deficient alkene such as acrylonitrile smoothly participated in the four-component coupling (**5n**). Nonactivated alkene and aryl

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alkene show no reactivity in the coupling, which may be attributed to radical polarity mismatch (**5o** and **5p**).

Encouraged by the fruitful compatibility and functional group tolerance of arylboronic acids and 1,3-enynes, a variety of alkyl radical precursors were investigated (Scheme 5). A

Scheme 5. Scope of Alkyl Radical Precursors[a]

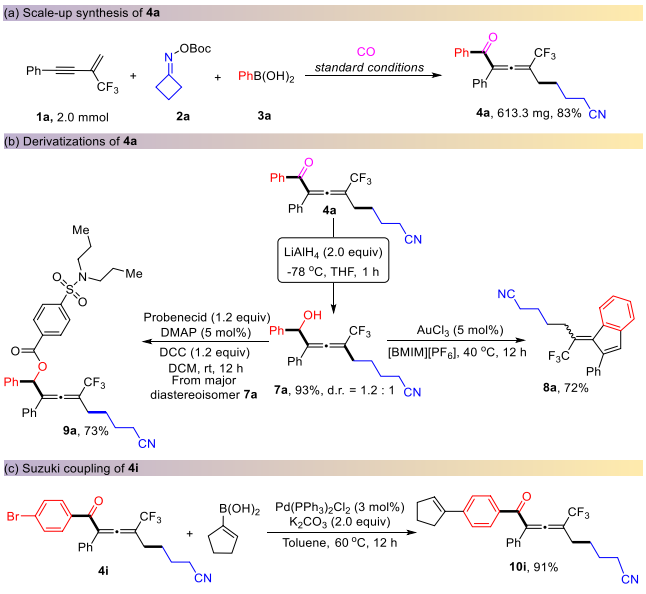


[a]Reaction conditions: **1a** (0.1 mmol), **2** or **2′** (0.2 mmol), **3** (0.2 mmol), $Ni(PPh_3)_2Br_2$ (5 mol %), dtbpy (5 mol %), Et_3N (0.3 mmol, 3.0 equiv), and H_2O (0.5 mmol, 5.0 equiv) under 1.0 atm of CO in 1,4-dioxane (1.5 mL) at 70 °C for 12 h. [b]**2′** instead of **2** and without H_2O.

range of cyclobutanone oxime esters decorated with phenyl, benzyl, ester, and amine proceeded efficiently, delivering the desired products **6a**−**6e** in 70−95% yields. The 3-oxetanone-derived substrate also provided the desired product **6f** in 72% yield. Unfortunately, the five-membered oxime ester was inert for this reaction. It is well-known that the alkoxy radical-mediated C−C bond cleavage is also a powerful strategy for the generation of the keto-alkyl radicals.[14] We were pleased to find that various cycloalkyl silyl peroxides are compatible with this four-component carbonylative reaction after slight modification of the standard conditions. In particular, less strained five-, six-, seven-, and 12-membered substrates were efficient for this transformation and afforded the desired allenyl ketones **6g**-**6l** in moderate yields. The high regioselectivity of the C−C bond cleavage is attributed to the favorable formation of more stable radicals (**6m** and **6n**).

To demonstrate the application potential of this four-component coupling protocol, the scale-up synthesis and the derivatizations were carried out, respectively (Scheme 6). When the model reaction was scaled up to 2.0 mmol, an 83% yield of **4a** was still obtained. Treatment of CF_3-allenyl ketone **4a** with $LiAlH_4$ at −78 °C afforded valuable CF_3-allenol **7a** in 93% isolated yield. CF_3-allenol **7a** could undergo the Au-catalyzed cascade hydroxyl elimination and cyclization to give product **8a** in 72% yield. Allenol **7a** could also be efficiently esterified with probenecid (a drug molecule) to afford ester **9a** in 73% yield. In addition, the bromo-containing product **4i**

Scheme 6. Scale-up Synthesis and Derivatizations of Products



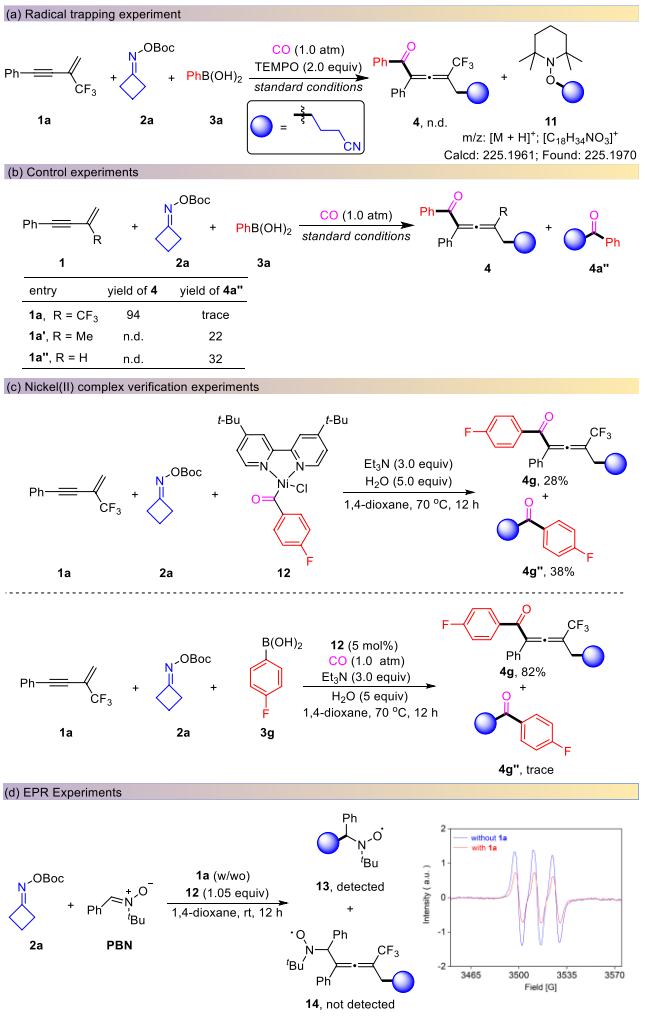
could undergo palladium-catalyzed Suzuki coupling with 1-cyclopentenylboronic acid to afford product **10i** in 91% yield.

To gain further insight into this nickel-catalyzed four-component coupling, several control experiments were carried out to elucidate the mechanism. First, a radical trapping experiment was performed by adding 2.0 equiv of TEMPO to the model reaction. The reaction was completely inhibited, and adduct **11** was detected by HRMS, indicating that the reaction proceeded via an alkyl radical pathway (Scheme 7a). Replacement of electron-withdrawing group CF_3 with electron-donating group CH_3 or H resulted in no formation of four-component product **4**. And the three-component product **4a′′** was isolated in 22% and 32% yields, respectively. These results showed that polarity matching is very important, making the addition of the alkyl radical to 1,3-enyne preferable to the capture of the alkyl radical by nickel (Scheme 7b). To further elucidate the mechanism, arylacyl nickel complex **12**[4b,15] was prepared in advance and then subjected to the reaction of **1a** and **2a**. In this case, the desired product **4g** was obtained in 28% yield, and aryl ketone **4g′′** was isolated in 38% yield (Scheme 7c). In addition, when 5 mol % of arylacyl nickel complex **12** was used as a catalyst instead of the combination of $Ni(PPh_3)_2Br_2$/dtbpy, the reaction still proceeded efficiently, and the target product **4g** was obtained in 82% yield. To further verify the role of the $Ar(CO)Ni^{II}$ complex, the electron paramagnetic resonance (EPR) study was performed (for details, see the SI). It was found that after stirring $Ar(CO)Ni^{II}$ **12**, cyclobutanone oxime ester **2a**, 1,3-enyne **1a** (w/wo), and the spin trapping agent phenyl *tert*-butyl nitone (PBN), the EPR signal of **13** was observed. These results suggest that the $Ar(CO)Ni^{II}$ **12** can undergo a single electron transfer (SET) event with cyclobutanone oxime ester **2a** to form the alkyl radical. These results indicate that $Ar(CO)Ni^{II}$ is the key intermediate for this transformation (Scheme 7d).

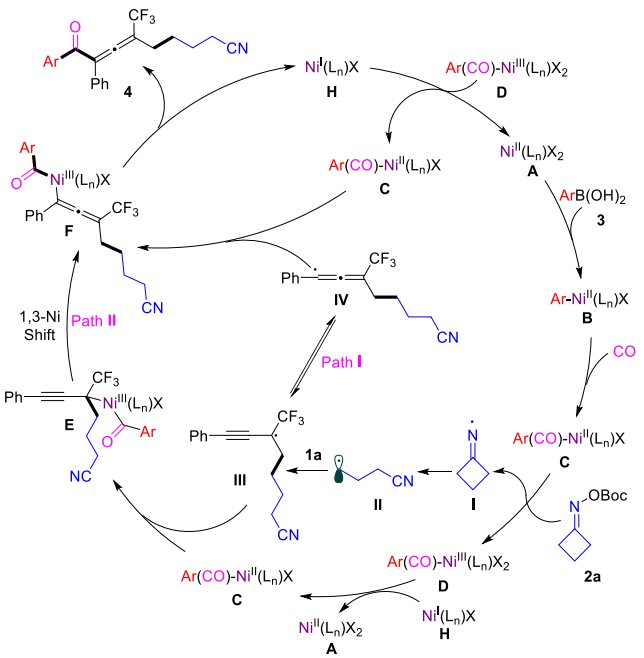
Based on the above experimental results and literature,[5,6,7h] a plausible mechanism was proposed for this nickel-catalyzed cascade (Scheme 8). The coupling begins with the transmetalation of Ni^{II} complex **A** with arylboronic acid **3** to yield

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Scheme 7. Mechanistic Studies



Scheme 8. Proposed Mechanism



aryl-Ni^{II} species **B**. Insertion of CO into aryl-Ni^{II} complex **B** gives acyl-Ni^{II} complex **C**, which undergoes a single electron transfer event with oxime ester **2a** to produce acyl-Ni^{III} complex **D** and iminyl radical **I**. Cyclic radical **I** suffers *β*-scission to give alkyl radical **II**. Subsequently, nucleophilic addition of alkyl radical **II** to the 1,3-enyne forms propargyl radical **III**, which tautomerizes to give allenyl radical **IV** (path **I**). Meanwhile, the comproportionation between acyl-Ni^{III} complex **D** and Ni^{I} species **H** occurs to yield Ni^{II} complexes **C** and **A**. Acyl-Ni^{II} species **C** reacts with allenyl radical **IV** to form allenyl-Ni^{III} complex **F**. Finally, reductive elimination of complex **F** produces product **4** and Ni^{I} species **H**. In addition, a 1,3-Ni shift pathway cannot be excluded. Here, the key allenyl-Ni^{III} complex **E** was formed by a 1,3-Ni shift of propargyl-Ni^{III} complex **E** (path **II**), which is generated by the interaction of radical **III** with acyl-Ni^{II} species **C**.

In conclusion, we have developed a novel Ni-catalyzed 1,4-alkylcarbonylation of 1,3-enynes under 1 atm of CO. To the best of our knowledge, this is the first example of four-component carbonylative coupling of 1,3-enynes. In this cascade, competitive three-component coupling and carbonylative coupling were successfully avoided by regulating the reactivity of oxime esters through leaving groups and controlling the polarity matching of alkyl radicals with 1,3-enynes. Preliminary mechanistic studies suggest that the acyl nickel complex is a key intermediate in this transformation. This one-pot protocol features mild conditions, broad substrate scope, and excellent functional group compatibility, thus providing a facile approach to the tetra-substituted CF_3-allenyl ketones. This modular synthetic strategy would broaden the application of carbonylation in organic synthesis.

■ ASSOCIATED CONTENT

Supporting Information

The Supporting Information is available free of charge at https://pubs.acs.org/doi/10.1021/acscatal.3c05776.

Experimental procedure and spectroscopic data for all compounds (PDF)

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Notes

The authors declare no competing financial interest.

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